MIL-F-18180D
30 May 1986
SUPERSEDING
MIL-F-18180C
30 April 1981

MILITARY SPECIFICATION

FLANGES AND FLANGED FITTINGS, PIPE, STEEL, CLASSES 150, 300, 400, 600, 900, 1500, AND 2500; GENERAL SPECIFICATION FOR

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers cast, forged, and deep drawn plate steel pipe flanges and cast and forged flanged fittings in the pressure-temperature ratings of class 250, 300, 400, 600, 900, 1500 and 2500. Reducing flanges without hubs and blind flanges may be manufactured from steel plate.

1.2 Classification. Flanges and flanged fittings shall be of the following types, styles and forms and shall be designated by the applicable definitive Military Specification Sheet part number (see 6.2(b), 6.3, and applicable Military Specification Sheet).

Type I - Flanges:

Style A - Welding Neck	(MIL-F-18280/1)
Style B - Socket-Welding	(MIL-F-18180/2)
Style C - Slip-On-Welding	(MIL-F-18180/3)
Style D - Lapped	(MIL-F-18180/4)
Style E - Threaded	(MIL-F-18180/5)
Style F - Blind	(MIL-F-18180/6)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4730

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

AMSC N/A

Type II - Flanged Fittings:

Style A - Elbow

Form 1 - 45 Degrees	(MIL-F-18180/7)
Form 2 - 90 Degrees	(MIL-F-18180/7)
Form 3 - 90 Degree Base	(MIL-F-18180/8)
Form 4 - 90 Degree Long Radius	(MIL-F-18180/7)
Form 5 - 90 Degree With Side Outlet	(MIL-F-18180/9)

Style B - Tee

Form 1 - Regular	(MIL-F-18180/10)
Form 2 - Base	(MIL-F-18180/11)
Form 3 - Side Outlet	(MIL-F-18180/12)

Style C - Cross

Form 1 - Regular	(MIL-F-18180/13)
Form 2 - Side Outlet	(MIL-F-18180/14)

Style D - Lateral

Form 1 - 45 Degrees	(MIL-F-18180/15)

Style E - Reducer

Form 1 - Concentric	(MIL-F-18180/16)
Form 2 - Eccentric	(MIL-F-18180/17)

Style F - Wye

Form 1 - True Wye (Y)	(MIL-F-18180/18)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-V-3 - Valves, Fittings and Flanges (Except for Systems Indicated Herein), Packaging of.

(See Supplement 1A for list of associated specification sheets.)

STANDARD

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as Department of Defense (DoD) adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A570 - Specification for Hot-Rolled Carbon Steel Sheet and Strip, Structural Quality.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section VIII, Divison I - Pressure Vessels
ANSI B16.5 - Steel Pipe Flanges and Flanged Fittings.

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern.

3.2 Standard commercial product. The flanges and flanged fittings shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard product. Additional features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard product shall be included in the flanges and flanged fittings being furnished.

3.3 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.4 Design. Flanges and flanged fittings shall be in accordance with ANSI B16.5.

3.4.1 Alternate blind and reducing flange design. Blind flanges and reducing flanges without hubs may be made from steel plate compositions listed in ANSI B16.5 under the required material group number.

3.4.2 Alternate "convoluted" flange design. Pipe flanges may be of the "convoluted" design. The outside diameter, bolthole circle diameter, quantity of boltholes, bolthole diameter, bore diameter of lapped, slip-on-welding, socket-welding and welding neck flanges, socket bore diameter of socket-welding flanges and hub diameter at butt weld end of welding neck flanges on the "convoluted" design shall conform to ANSI B16.5 dimensions for the applicable pressure class and nominal pipe size. Any void areas in the "convoluted" flange structure shall contain at least two ventilating/drain holes equally spaced on the bolthole circle diameter. When material group number 1.4 is specified, "convoluted" flanges shall be manufactured from ASTM A570 Grade 45 steel or a material composition listed under material group number 1.4 in ANSI B16.5.

3.4.3 Boltholes. All flanges or flanged fittings shall be furnished with ANSI B16.5 boltholes.

3.4.4 Bore diameter (waterway). The waterway bore diameter for socket-welding or welding neck flanges shall equal the inside diameter of specified schedule pipe (see 6.2(b) and applicable Military Specification Sheet).

3.4.5 Facings. The facing for flanges or flanged ends of flanged fittings shall be in accordance with one of the ANSI B16.5 patterns, as specified (see 6.2(b) and applicable Military Specification Sheet). "Convoluted" flange flat facing shall be acceptable for either flat or raised face requirements.

3.4.6 Pressure-temperature rating. The pressure-temperature rating of flanges or flanged fittings shall be in accordance with ANSI B16.5 Class 150, 300, 400, 600, 900, 1500 and 2500, as specified in (see 6.2(b) and applicable Military Specification Sheet).

3.4.7 Size. The size of flanges or flanged fittings shall be ANSI B16.5 nominal pipe size (in inches), as specified (see 6.2(b) and applicable Military Specification Sheet).

3.5 Auxiliary connections (flanged fittings). Flanged fitting auxiliary connections shall be ANSI B16.5 sizes, kinds, and locations, as specified (see 6.2(b) and applicable Military Specification Sheet).

3.6 Protective finish. All flanges and flanged fittings, except for stainless steel flanges and flanged fittings, shall be coated with manufacturer's standard commercial paint or rust inhibitor. Stainless steel flanges and flanged fittings shall have a passivated finish.

3.7 Pressure tests.

3.7.1 ANSI B16.5 flanges. Cast ANSI B16.5 flanges shall be hydrostatically tested as specified in 4.4.1. All other ANSI B16.5 flanges are not required to be hydrostatically tested.

3.7.2 "Convoluted" flanges. "Convoluted" flanges shall be pressure-proof tested as specified in 4.4.2.

3.7.3 ANSI B16.5 flanged fittings. ANSI B16.5 flanged fittings shall be hydrostatically tested as specified in 4.4.1.

3.8 Marking. Flanges and flanged fittings shall be marked in accordance with ANSI B16.5.

3.9 Workmanship. Workmanship shall be in accordance with the commercial practice of the leading manufacturers of flanges or flanged fittings. Machined surfaces shall have no burrs. Nonmachined surfaces, such as cast and forged surfaces and bar stock flats, shall have no fins, cracks or spongy area. Flanges and flanged fittings shall be free from detrimental grooving, indentations, flaws, and slivers which would interfere in their use and shall be free from sand, dirt, corrosion and other harmful extraneous material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise

specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105. The unit of product shall be one complete flange or flanged fitting. All items of the same classification offered for delivery at one time shall be considered a lot for the purpose of inspection. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for a complete reinspection. Resubmitted lots shall be reinspected using tightened inspection. If the rejected lot was screened, reinspection shall be limited to the defect causing rejection. If the lot was reprocessed, reinspection shall be performed for all defects. Rejected lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.2.1 Sampling for examination. Examination shall be based on inspection level II and an Acceptable Quality Level (AQL) of 4.0 percent defective.

4.2.2 Sampling for tests. Tests shall be based on inspection level S-4 and an AQL of 4.0 percent defective.

4.3 Examination. Each flange and flange fittings shall be examined for defects listed in table I. Each attribute within each classification of multiple defects shall constitute one defect.

TABLE I. Classification of defects.

Classification	Defects	Requirement paragraph
Major:		
101	Flanges and flanged fittings not as specified.	1.2 and 3.1
102	Material composition not as specified. Not one of those allowed in specified material group number.	3.3
103	Design not as specified.	3.4.1 and 3.4.2
104	Boltholes not cast, drilled or	3.4.3
105	Bore dimensions not as specified.	3.4.4
106	Facings not as specified.	3.4.5
107	Size not as specified.	3.4.7

TABLE I. Classification of defects. (continued)

Classification	Defects	Requirement paragraph
Minor:		
201	Finish not as required.	3.6
202	Improper markings.	3.8
203	Poor workmanship. Surfaces with imperfections or corrosive material.	3.9

4.4 Tests.

4.4.1 Hydrostatic test. The hydrostatic test for flanged fittings and applicable flanges shall be no less than 1.5 times the 100 degrees Fahrenheit (°F) pressure-temperature rating for the pressure class of the flange or flanged fitting rounded off to the next higher 25 pound-force per square inch increment. The test shall be made with water, which may contain a corrosion inhibitor, with kerosene, or with another suitable fluid provided its viscosity is no greater than that of water, at a test temperature not above 125°F. The test duration shall be a minimum of 1 minute for 2 inch nominal pipe size and 3 minutes for 10 inch nominal pipe size and larger. No visible leakage is permitted through the pressure boundry walls or at flange connection area (flange facing).

4.4.2 Pressure proof test. The pressure proof test shall be in accordance with paragraph UG-101 in Section VIII, Division 1 of the ASME Boiler and Pressure Vessel Code. No visible leakage is permitted through the pressure boundry walls or at the flange connection area (flange facing) and the flange shall not be damaged, so flange cannot be used again without repairing, except for the burst pressure test.

4.5 Packaging inspection. The inspection of the preservation, packing, and marking shall be in accordance with the requirements of section 4 of MIL-V-3.

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be in accordance with the requirements of MIL-V-3 with the level of preservation and the level of packing as specified (see 6.2).

6. NOTES

6.1 Intended use. Flanges and flanged fittings covered by this specification are intended for use in low through high pressure piping systems conveying steam, water, oil, or gas.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Military Specification Sheet definitive part number required (see 1.2, 3.4.4, 3.4.5, 3.4.6, 3.4.7, 3.5, 6.3 and applicable Military Specification Sheet).
c. Level of preservation and level of packing required (see 5.1).

6.3 Definitive Military Specification Sheet part number. The Military Specification sheet part number is a definitive part number which corresponds to the type, style, and form of flanges or flanged fittings covered by this specification and defines the requirements of the options presented under this specification (see 6.2b). The exact format of the Military Specification Sheet part number for a particular flange or flanged fitting is presented on its Military Specification Sheet.

6.4 Subject term (keyword) listing.

Alternate
Auxiliary
Boltholes
Bore Diameter
Convoluted
Design
Facing
Fittings
Flanges
Hydrostatic
Materials
Pipe
Pressure Test
Proof
Protective

6.5 Supersession data. Under classification (paragraph 1.2); type II, style A, forms 6 and 7 of MIL-F-18180B were deleted in MIL-F-18180C and type II, style D, form 2 was changed to type II, style F, form in MIL-F-18180C. They are unchanged in MIL-F-18180D.

Custodians:

Army - ME
Navy - YD

Review Activity:

DLA - CS

User Activities:

Navy - CG, MC

Preparing Activity:

Navy - YD

(Project 4730-0703)

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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Commanding Officer (156)
Naval Construction Battalion Center
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-18180	FLANGES AND FLANGED FITTINGS, PIPE, STEEL, CLASSES 150, 300, 400, 600, 900, 1500, AND 2500

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426

PREVIOUS EDITION IS OBSOLETE.